

Title

The Complexities of Train Schedule Management: A Look at NS's Planning Process and the Push for Optimization

Abstract

Train schedules are a crucial part of our daily lives, but have you ever thought about the complexity behind creating them? From ensuring seats are available to deploying personnel, there are numerous factors to consider. Despite significant advancements in planning automation, there is still room for improvement, especially in hub planning optimization. So, what's happening under the hood at NS, and why is optimizing railway planning so challenging? We interviewed Bob Huisman, Manager of Research and Development Hub Logistics at NS, to learn more about how AI is revolutionizing the train scheduling management process.

Bio

Bob Huisman is a respected figure in the railway industry, with his career spanning several decades. Huisman currently holds the position of Manager Research & Development Hub Logistics at NS, where he is responsible for delivering innovative methods and tools for planning and scheduling shunting-related processes at railway hubs, as well as assessing the logistic process capacity of railway hubs. Beyond the technical and scientific aspects of his work, Huisman sees it as an opportunity to make a social contribution. He describes himself as 'having one foot firmly planted in the business world and the other in academia'. His career path is a testament to his ability to bridge the gap between research, development, creativity, and challenging problems. Huisman is one of the principal investigators of the LTP ROBUST programme and the chair of the users' committee.

Interview

'At first glance, the process of train schedule management may seem simple - travelers at the station, a train ready to board.' But as Huisman points out, it's actually a complex interplay of factors, while peering into the direction of Utrecht Central Station, one of the important hubs in the railway network. The hub planning, as part of the overall railway planning, involves ensuring trains are in the correct composition to maximize seat availability, that they arrive at the right platform at the right time, and that they are in good technical condition. Additionally, once a train reaches its final destination, it must be checked for technical issues, cleaned, potentially rearranged, and parked in a way that maximizes space efficiency.

But that's not all. Railway planning and control involve fleet assignment and the deployment of personnel, which comes with its own set of complicated and important limiting factors. Employment conditions, work variation, and the ability of colleagues to come home at the end of the day are just some of the factors that the team of Huisman must take into account. 'Train schedule management is almost paradoxical. As a traveler, you may experience that we have one timetable during the entirety of the year, but under the hood, the rail sector makes a unique plan for every single day of the year. This involves planning for the timetable, fleet and staff, as well as for the 34 hubs – the stations with connected yards where multiple train lines converge – months in advance.'

Over the past two decades, NS has experienced a significant increase in automation in network planning. However, there's still no automation in hub planning, which Huisman notes as remaining an obstacle to overcome. Currently, this daunting task is solely on the shoulders of human hub planners, who are responsible for what is called the "knitting process". 'The "knitting process" involves juggling a multitude of factors simultaneously and making decisions in real-time, ensuring that passengers arrive at their destinations safely and smoothly.'

How do those train-scheduling experts manage to make everything run like clockwork?

'It is a multi-step process. First, we create a timetable, then assign our fleet, and lastly assign our colleagues.' Albeit the largest, NS is only one of the 7 Dutch operators on the Dutch rail network, with ProRail responsible for infrastructure management, capacity allocation, and traffic control. Huisman notes that planning NS's train operation involves multiple iterations before arriving at the final basic pattern, which is then translated into a blueprint for each weekday, and subsequently each specific day. The schedule is finalized a month in advance for a specific calendar day, and from then on, NS can still make adjustments up to two days prior. 'Once the plans are handed over for operational execution, controllers at ProRail and NS must act quickly in the event of a collision, malfunction, or employee illness. At that moment, it resembles tinkering more than actual planning by optimization', Huisman notes.

NS has been making significant progress since 2015 towards automating the hub planning, but Huisman emphasizes that there are many "dirty details" that need to be taken into account. One of the most gratifying moments of this project came in 2020, just before the onset of the pandemic. 'We were able to demonstrate based on a proof of concept that we are going to make it', Huisman recalls. 'Moments like these, when colleagues have confidence in the success of a project and more resources become available, those keep me young. I have resolved that when I retire, to have this project standing,' he says with a sense of pride. 'This project represents a unique collaboration with young researchers, a continuous flow of PhD and master students, and has been one of the most personally rewarding projects of my career.'

Why is automation and optimization so important?

'Two goals for the rail system are difficult to reconcile: on the one hand, meeting a growing demand for transport, and, on the other hand, a robust operation. The driving force for automated hub-planning support is the need to fix the plans as late as possible and be able to make changes online. That will improve the robustness of the transportation process, rail infrastructure usage, and seat availability,' Huisman notes.

Currently, to anticipate uncertainties and unforeseen disruptions, slack is incorporated into the planning, with respect to both space and time. This slack allows railroad planners to be flexible and deal with details that will become clear on the day of operation itself; it gives them space to breathe when things go wrong. 'Reducing slack to facilitate future passenger volumes increases the risk of a domino effect of disruptions; however, fast automated support for planning and

control may compensate for this. All of these challenges beg the question: how do we achieve sustainable growth, act more dynamically, and be more robust, while using our existing resources more efficiently?

NS seems to already have invested considerable resources into research that helps professional planners create more optimized plans in less time. Why is this process so tough?

'Good question. Now, picture the entire rail system as a massive, interconnected wirework - a complex maze that requires meticulous planning to operate smoothly. Countless variables can impact the system, making optimization a daunting task. It's not just a matter of flipping a few switches, pushing a few buttons, and moving a few trains around''.

'Hub planning is a combinatorial problem with an enormous search space in which it is hard to find good and feasible solutions. Moreover, the need to fix plans as late as possible requires modeling many dirty details of the real world and complex safety rules, which excludes linear optimization methods. 'When we started in 2015, hub planning had been the topic for a well-known international competition in the field of Operation Research. Curious as we were, we reached out to all the competition's prize winners to see if they knew something we didn't. Eventually, we concluded that no practical solution had been found yet, which was why we decided to set up a long-term research and development program ourselves. One way or another, we had to find a way. Together with our academic partners, we finally succeeded in building a working system, mainly based on powerful local search, combined with other methods like linear optimization and constraint programming. We coined it the Hybrid Integrated Planning method (HIP). Although the system generates plans that are acceptable to professional planners, continuation of the research is needed to enhance the system's functionality.'

'Following the progress made by DeepMind, preceding AlphaZero, we started a research track specifically focused on applying deep-reinforcement learning to multi-agent pathfinding. The idea was to build up a logistics plan by generating a chain of individual actions, representing train shunting movements. After six years of research together with our academic partners, we found that brute force local search still outperformed our various complex reinforcement learning approaches, even on simplified models of reality. Taking a step back to see the forest for the trees, we halted the program and shifted our focus.

Our current research direction is aimed at using machine learning to complement local search, constrained programming, and linear optimization. The challenge is to find and modify plans more quickly, specifically plans that are easily understood by humans. To build a system that exhibits some intelligent behavior in the future, it must be able to learn from previous situations and to communicate at some abstract level with its users. We still have a long way to go, which asks for perseverance and creativity', Huisman notes.

“Humans in the loop” does not conform to the traditional view on automation, in which AI systems entirely replace human operators. However, recently, researchers have started to view automation as a two-dimensional process, where humans and machines work together and complement each other’s strengths and weaknesses to achieve a common goal.

‘In our unique use case, we are looking for automatic tools to support people’, Huisman emphasized. ‘Hub planning is a challenge that humans are quite good at and a task that requires substantial creativity, ingenuity, and the ability to color outside the lines - but it takes a lot of time. On the other hand, algorithms can speed up the planning processes, but currently cannot handle their full complexity. That is why we are focused on creating systems that help people to adjust the “knitting process” more efficiently; reducing slack and maximizing space usage to meet the growing demand for train travel.’

When it comes to AI, Huisman has a unique perspective. ‘AI research can bring us closer to understanding human intelligence, yes. However, as long as we don’t understand and have not defined human intelligence, stop talking about artificial intelligence that has to replace humans.’ Instead, Huisman believes that we should focus on building super tools that exhibit intelligent behavior, regardless of whether there’s a steam engine or neural network powering it. ‘I see neural networks and reinforcement learning as ingredients, among others, to create value’, Huisman explained, adding that it’s all about developing an overall system that can deal with the intricacies of logistic planning in cooperation with humans. ‘AI has the potential to disrupt the approach we take in this, but it is not just about generating plans automatically; you have to communicate the output to planners and make it understandable to them, give humans control over plan qualities, and link the output to the systems of other parties.’

Recently, ICAI announced the LTP ROBUST program, a new initiative supported by the University of Amsterdam and 51 partners from government, industry, and academia. As part of the program, 17 new AI labs are established that focus on the development of trustworthy AI technology to address socially relevant issues in areas such as healthcare, logistics, media, food, and energy. Can you elaborate on the importance of trust in AI systems and your role in the program?

‘Trust is a crucial factor in the adoption of AI systems. Our perspective is that the public, customers, travelers, patients, users, and authorities all base their judgment not only on the functionality of the AI algorithm in isolation. Rather, they base it on the whole of interacting ICT, the organization behind it, the procedures put in place to regulate it, the UI, and the availability of the system. Trustworthiness of an individual AI algorithm is a necessary, but not a sufficient condition, for its effective use in a system. Therefore, research into creating such AI systems necessitates a symbiotic relationship between academia and industry. In the end, private or governmental organizations set the specifications of the system, design and build the system, and operate the system over the years. In LTP, research, development, and system engineering meet to obtain social impact by operational trustworthy systems.

Trust is also contextual and domain-specific; the risks of a medical diagnosis differ from the risks of logistics planning or music recommendation, and people rely on different systems in different ways. The program's approach is to start with a system vision and a targeted research question for each lab, with the private partners playing a vital role in validating the output and asking the right questions. 'While the validation and questioning may vary for different fields, the general approach to winning the trust of the user can be similar. As one of the principal investigators and the chairman of the overarching user committee, my role is to oversee the cooperation between the partners and ensure knowledge transfer between labs.'

The RAIL Lab, a collaboration between Delft University of Technology, Utrecht University, ProRail, and NS, is one of the LTP ROBUST Labs joining ICAI. Its goal? Working towards algorithmic support to ensure safe and reliable logistic operations and capacity planning that is trusted by human experts. Explainable AI plays a role in this.

'Explainability is often seen from the research world as: if I could just explain why my algorithm came to this conclusion, and if I change something about my input, how would my evaluation change? It's almost an internal accountability from your algorithm to the outside world, which is necessary, but might only be sufficient to accept or reject an individual prediction of the system. The question is whether that is sufficient for humans to use and accept the system as a sparring partner for decision support.'

Huisman emphasized the importance of setting standards for what an explanation of an algorithm should look like. 'Authorities often require a deeper understanding of how an algorithm works, including how it makes considerations, what information it looks at, what information is necessary to make a good choice, and how uncertain the algorithm is in its output. Furthermore, for specific instances, humans may ask counterfactual questions to understand why one decision is proposed and not another. By understanding the requirements for human decision-making, we can create more effective explanations that provide a more complete understanding of the algorithm's decision-making process. Since the user is often responsible for the final decision to be taken, he wants to be sure it is the right one.'

To address these challenges, each LTP ROBUST lab will include a researcher with a background in social and behavioral science. The RAIL Lab is a testament to this effort, with one PhD student focusing on the cooperation between human and AI planners. This study will reveal requirements, expectations, and potential pitfalls of human-AI interaction, specifically of interaction with algorithmic planners. These results will be augmented with data science techniques to extract important factors from past decision-making and planning processes, to develop a computational cognitive model of the decision and planning process.

Huisman sees a colorful future ahead: 'NS has its fair share of critics - some say it's too big, bureaucratic, or slow. On the other hand, I know of few other companies that have invested as much time and resources into innovative projects like railway planning, as NS has in the Netherlands.' Optimizing rail systems is a complex task that will require many more years of research and a delicate balance between human expertise and advanced AI algorithms.

However, Huisman and his colleagues are committed and up for the challenge. 'With LTP ROBUST and RAIL Lab's ongoing efforts, we can hope to see more trustworthy, efficient, and seamless rail systems in the near future.'

We hope that through this interview, you learned a bit more about NS and the intricacies of railway planning. NS, ProRail, and their academic partners, TU Delft and the University of Utrecht, are currently recruiting PhD students for the RAIL Lab. If you are interested in a complex technical AI challenge, in the light of a social contribution, check out their webpage: <https://icai.ai/icai-labs/rail/>. The next time you're waiting for your train, take a moment to appreciate the intricate dance of 22.000 employees that's happening behind the scenes to help you get to your destination, and maybe consider joining us!